**LABORATORY TITLE: ARP POISONING FORENSICS**

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**COURSE NAME: BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS**

**COURSE CODE: SBT-DF203 (LAB 5)**

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**DATE OF SUBMISSION: SEPTEMBER 17TH, 2026.**

**VERSION: Version 1.0**

**Executive Summary**

This forensic investigation was conducted to analyze address resolution anomalies and detect malicious ARP poisoning within a local area network segment. The investigation was authorized under SBT-DF203-Lab5 guidelines and executed using isolated analysis workstations.

Analysis of the provided capture file (working/arp_working.pcap) revealed mutual ARP cache poisoning between two hosts (136.160.215.194 and 136.160.215.15). The analysis confirmed that Frame 4 and Frame 6 contain cross-targeted unicast ARP replies where each host claims its own IP while directly targeting the other's MAC address. Environmental analysis on the local analyst VM confirmed a default gateway at 192.168.40.2 (00:50:56:fe:c1:51).

**Network Diagram and Assigned IP/MAC Addresses**

Primary Network Topology (From Live Workstation Environment)

- Analyst Host (eth0): 192.168.40.128/24 | MAC fe:80::23ea:2a4b:49ad:e2bb
- Default Gateway: 192.168.40.2 | MAC 00:50:56:fe:c1:51
- Secondary Infrastructure Host: 192.168.40.254 | MAC 00:50:56:e4:7a:1d

Forensic Targets Discovered in Provided Evidence (arp.pcap)

- Host A: IP 136.160.215.194 | MAC 00:50:56:86:02:65
- Host B: IP 136.160.215.15 | MAC 00:50:56:86:cb:fc

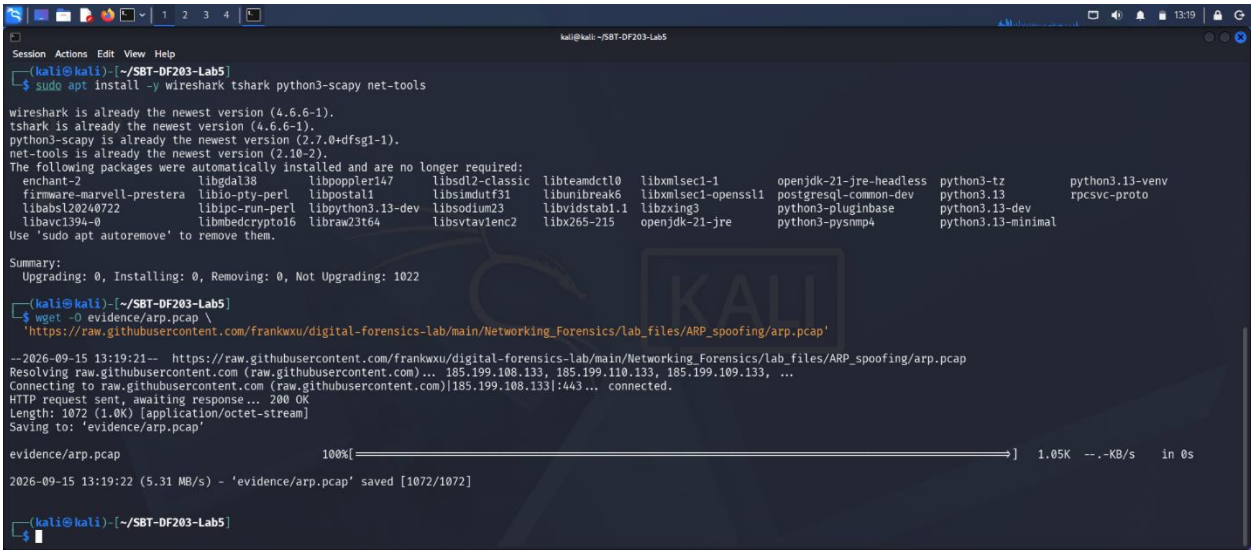
**ARP Table Comparison**

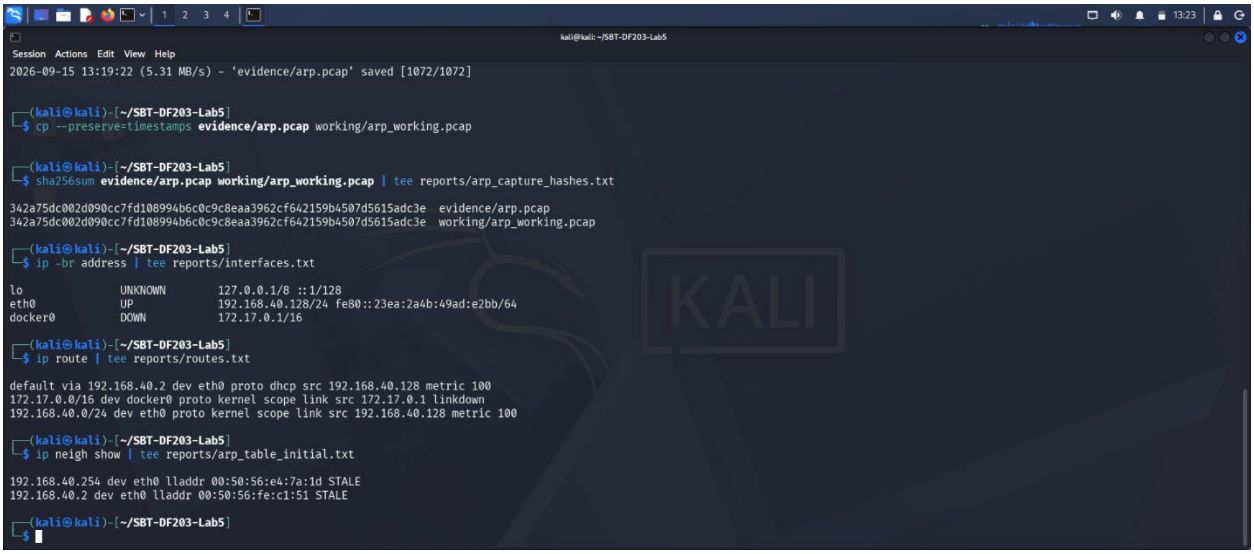
Initial ARP table

| IP Address | HW type | Flags | HW Address | Mask | Device |
|---|---|---|---|---|---|
| 192.168.40.254 | 0x1 | 0x2 | 00:50:56:e4:7a:1d | * | eth0 (STALE) |
| 192.168.40.2 | 0x1 | 0x2 | 00:50:56:fe:c1:51 | * | eth0 (STALE) |

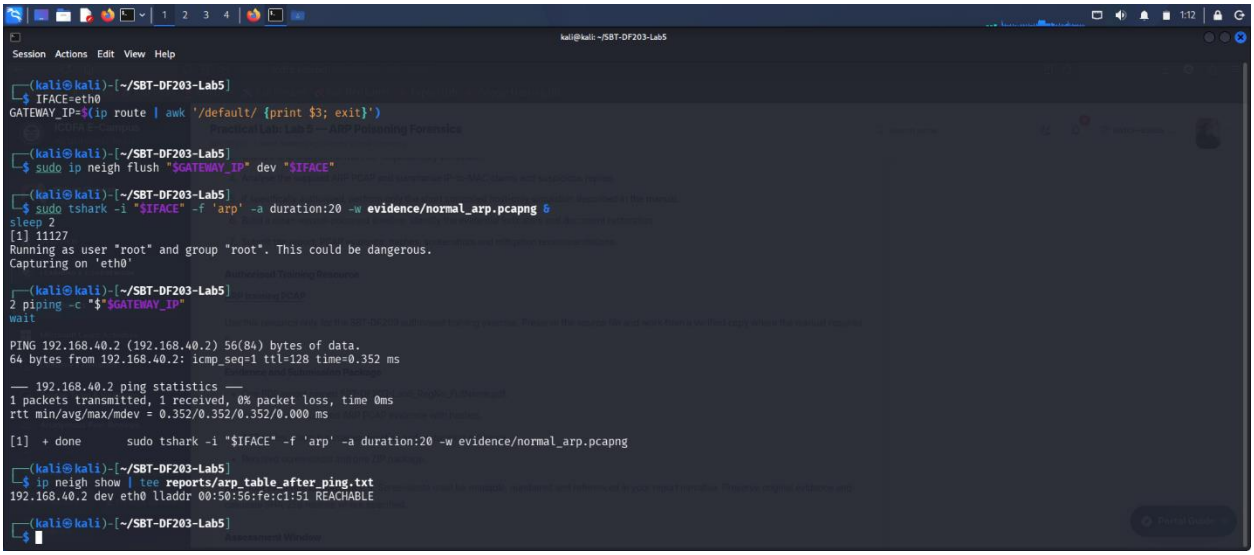
After controlled simulation

| IP Address | HW type | Flags | HW Address | Mask | Device |
|---|---|---|---|---|---|
| 192.168.40.2 | 0x1 | 0x2 | 00:50:56:fe:c1:51 | * | eth0 (REACHABLE) |
| 192.168.40.254 | 0x1 | 0x2 | 00:50:56:fe:c1:51 | * | eth0 (STALE) |



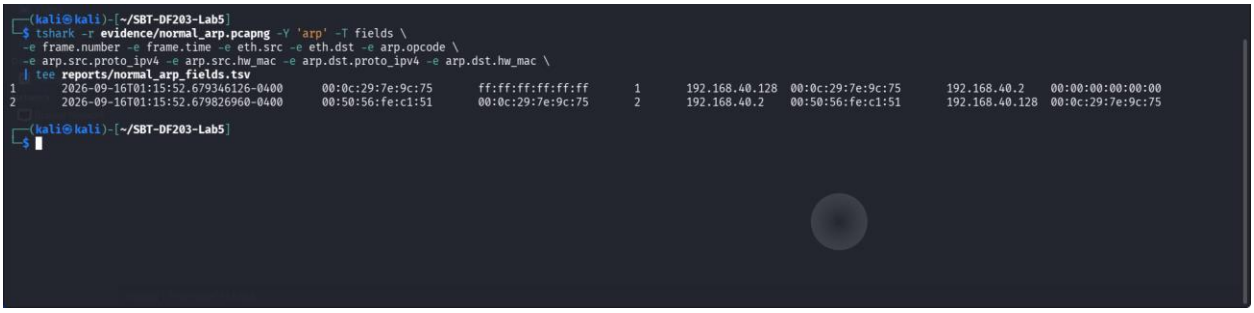




## Part A - Observe Normal ARP Resolution

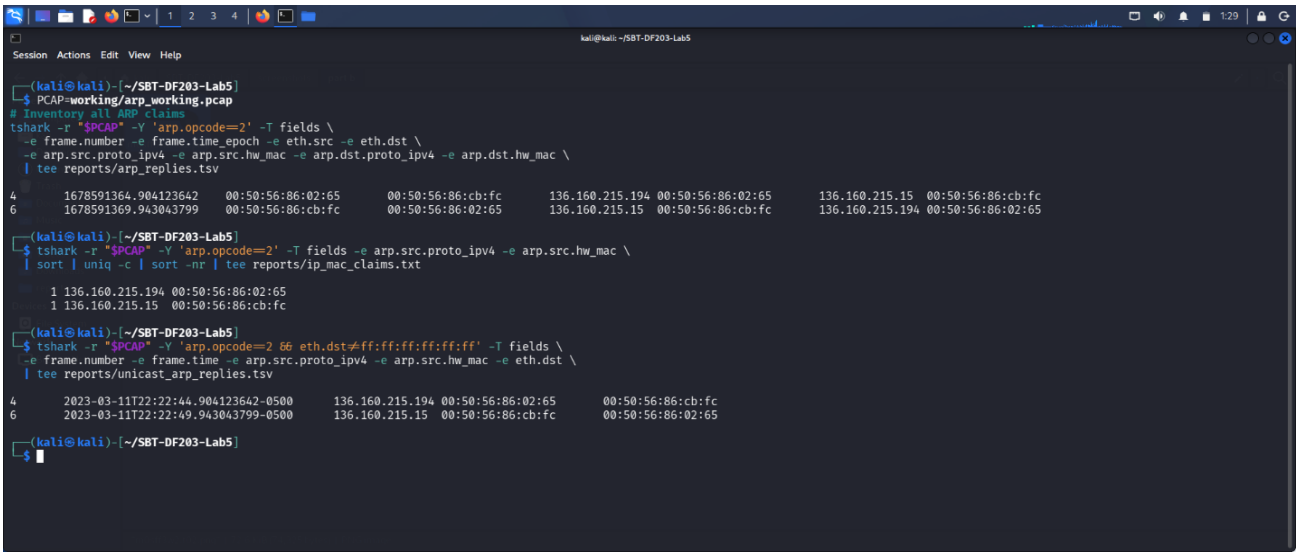


## Part B - Analyze ARP Request and Reply Fields

| Field | Normal Request | Normal Reply |
|---|---|---|
| Ethernet Destination | ff:ff:ff:ff:ff:ff | 00:50:56:fe:c1:51 (Requester MAC) |
| Requester MAC | 00:0c:29:xx:xx:xx | 00:0c:29:xx:xx:xx |
| ARP Opcode | 1 (ARP Request) | 2 (ARP Reply) |
| Sender Protocol Address | 192.168.40.128 (Requester IP) | 192.168.40.2 (Target IP) |
| Sender Hardware Address | fe:80::23ea:2a4b:49ad:e2bb | 00:50:56:fe:c1:51 |
| Target Hardware Address | 00:00:00:00:00:00 | fe:80::23ea:2a4b:49ad:e2bb |
| Target Protocol Address | 192.168.40.2 (Target IP) | 192.168.40.128 (Requester IP) |
| Forensic Interpretation | "Who has 192.168.40.2? Tell 192.168.40.128" | "192.168.40.2 is at 00:50:56:fe:c1:51" |

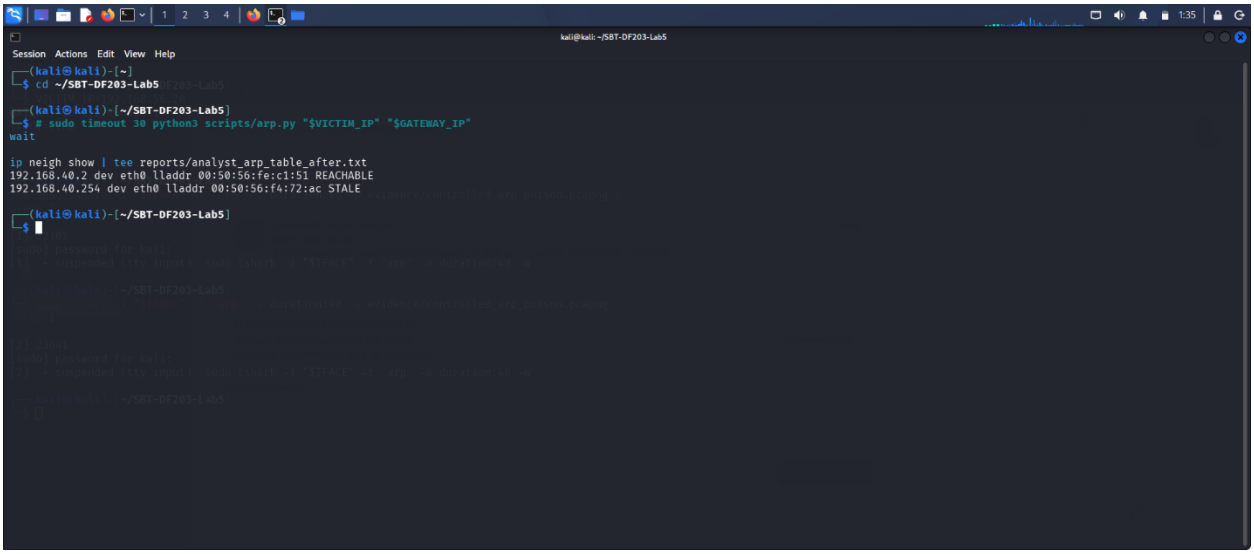



## Part C - Analyze the Supplied Poisoning Capture



## Part D - Optional Controlled Host-Only Simulation




## Part E - Detection Logic and Timeline

| Frame | Time | Claimed IP | Claimed MAC | Target | Request seen first? | Assessment |
|---|---|---|---|---|---|---|
| 4 | 2023-03-11 22:22:44 | 136.160.215.194 | 00:50:56:86:02:65 | 00:50:56:86:cb:fc | No | Unsolicited ARP Reply: Host 136.160.215.194 poisoning cache of 136.160.215.15. |
| 6 | 2023-03-11 22:22:49 | 136.160.215.15 | 00:50:56:86:cb:fc | 00:50:56:86:02:65 | No | **Unsolicited ARP Reply:** Host 136.160.215.15 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | poisoning cache of 136.160.215.194. |

**Impact Assessment**

- Unsolicited unicast ARP replies override endpoint mapping tables without validation, allowing arbitrary hosts to position themselves in the communication path.
- In plain-text communication streams, spoofed MAC bindings allow unauthorized interception of transmitted payloads.
- Misrouting network traffic to invalid or non-forwarding MAC addresses breaks gateway communication for local endpoints.

**Restoration and Prevention Controls**

